



CERTIFICATE OF TESTS

Heat: 24102191
Material: 5L- X52 4-1/2X.337 (15.00#)BR.DRL.PEB (11000102)
Vendor: Big River Steel - Osceola AR US (180431)

MFG Date: 09/24/2024

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0600	1.0400	0.0100	0.0010	0.0400	0.0500	0.0900	0.0170	0.1800	0.0010	0.0030	0.1281
	Al: 0.0380		Sn: 0.0048		N: 0.0074		B: 0.0002		Ca: 0.0030		Nb: 0.0020	
Chem	0.0692	1.0691	0.0089	0.0011	0.0385	0.0434	0.0697	0.0130	0.1861	0.0009	0.0046	0.1367
	Al: 0.0365		Sn: 0.0041		N: 0.0196		B: 0.0001		Ca: 0.0021		Nb: 0.0019	
Chem	0.0687	1.0580	0.0091	0.0014	0.0378	0.0435	0.0669	0.0117	0.1801	0.0008	0.0044	0.1352
	Al: 0.0342		Sn: 0.0041		N: 0.0183		B: 0.0001		Ca: 0.0020		Nb: 0.0020	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 68.75 TS (ksi): 72.90 EL (%) : 30.00 HRB: 90.00 Y/T(%): 0.94

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	79.00	100.00
Test Temp (°F) : -20	2	83.00	100.00
Location : 3T90	3	80.00	100.00
	Average	80.67	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-24,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.


Quality Control Dept